Data Mining

Laboratory Report — Lab 3

Decision Tree Classification

Gini vs. Entropy | Holdout & Cross-Validation | Depth Analysis

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Course Information

Course	Data Mining
Lab Number	3
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Datasets:

German Credit	1000 instances, 24 numeric features, 2 classes
Waveform v1	5000 instances, 21 continuous features, 3 classes

Department of Computer Science

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1 Introduction

Decision Trees are one of the most interpretable supervised learning algorithms. They recursively partition the feature space by selecting the split that maximises an impurity criterion, producing a flowchart-like model that can be directly inspected.

This report evaluates decision tree classifiers on two benchmark datasets from the UCI Machine Learning Repository:

- **German Credit** (1 000 samples, 24 numeric features, binary label: 0 = good credit, 1 = bad credit; 70 %/30 % split).
- **Waveform** (5 000 samples, 21 continuous features, 3-class label: 0, 1, 2; balanced classes, each ≈ 33%).

Three evaluation protocols are studied:

Part I — Gini impurity.
Holdout (90/10 split, 5 repeated trials) and 10-fold cross-validation.

Part II — Entropy impurity.
Same evaluation protocol; results compared with Part I.

Part III — Tree depth.
Effect of `max_depth` on training and test accuracy using the holdout method.

All experiments use `scikit-learn`. Precision and F-measure are computed with *weighted* averaging to correctly handle class imbalance.

2 Part I — Decision Tree with Gini Impurity

2.1 German Credit Dataset

2.1.1 Holdout Method (90/10, 5 Trials)

The dataset is split 90 % training / 10 % test using a different random seed for each trial. A full-depth `DecisionTreeClassifier(criterion='gini')` is trained and evaluated on the held-out 10 % each time.

Table 1: German Credit — Holdout Results (Gini)

Trial	Accuracy	Precision	F-measure
Trial 1	0.6600	0.7044	0.6762
Trial 2	0.6700	0.6669	0.6684
Trial 3	0.6600	0.6491	0.6535
Trial 4	0.6500	0.6471	0.6485
Trial 5	0.6300	0.6499	0.6385
Average	**0.6540**	**0.6635**	**0.6570**

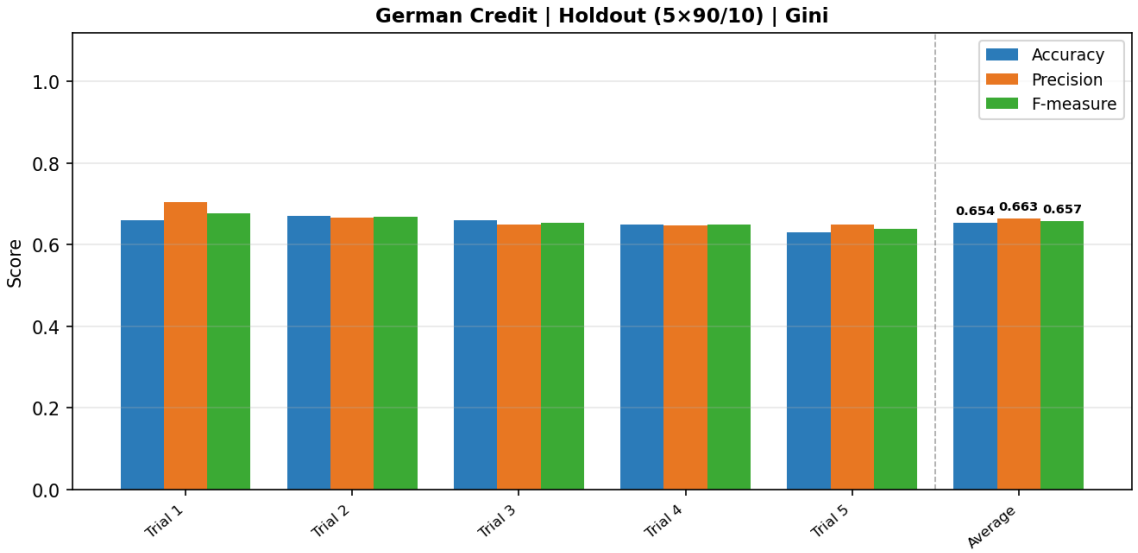


Figure 1: German Credit — Holdout metrics per trial (Gini)

2.1.2 10-Fold Cross-Validation

Table 2: German Credit — 10-Fold CV Results (Gini)

Fold	**Accuracy**	**Precision**	**F-measure**
Fold 1	0.6600	0.7044	0.6762
Fold 2	0.6300	0.6329	0.6314
Fold 3	0.7000	0.6936	0.6965
Fold 4	0.7300	0.7454	0.7342
Fold 5	0.7200	0.7141	0.7167
Fold 6	0.5900	0.6300	0.6058
Fold 7	0.7300	0.7215	0.7238
Fold 8	0.6700	0.6894	0.6781
Fold 9	0.6500	0.6832	0.6617
Fold 10	0.7200	0.7163	0.7178
Average	**0.6800**	**0.6931**	**0.6842**

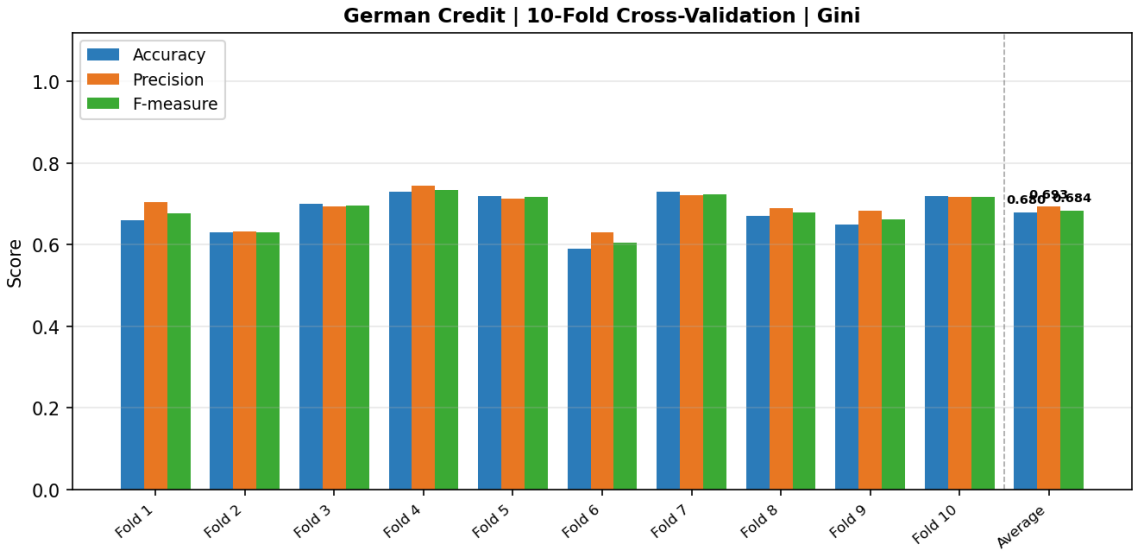


Figure 2: German Credit — 10-Fold CV metrics per fold (Gini)

2.2 Waveform Dataset

2.2.1 Holdout Method (90/10, 5 Trials)

Table 3: Waveform — Holdout Results (Gini)

Trial	Accuracy	Precision	F-measure
Trial 1	0.7440	0.7467	0.7438
Trial 2	0.7160	0.7180	0.7162
Trial 3	0.7760	0.7781	0.7763
Trial 4	0.7400	0.7405	0.7394
Trial 5	0.7680	0.7680	0.7679
Average	**0.7488**	**0.7503**	**0.7487**

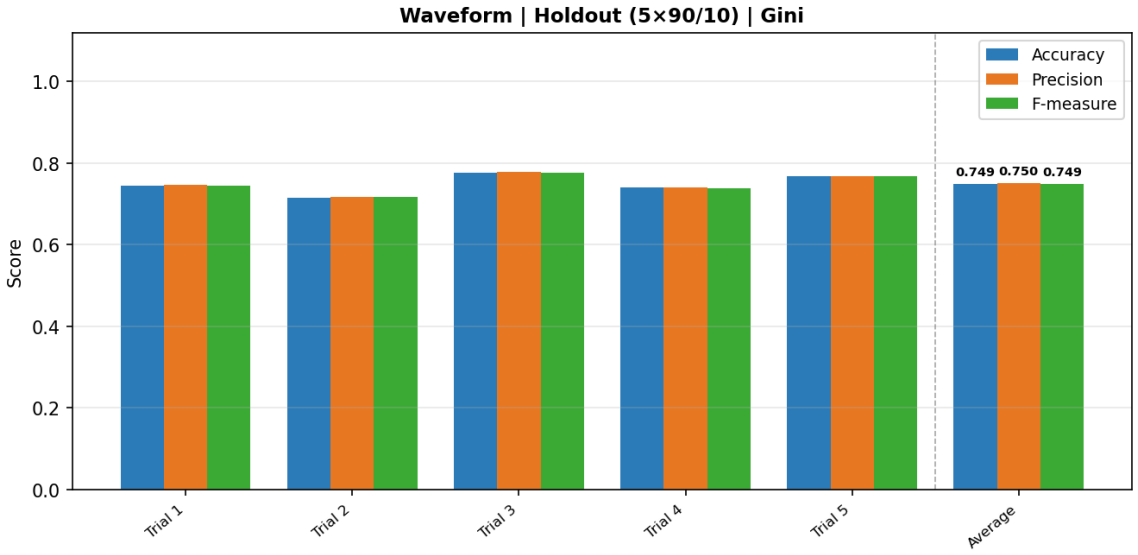


Figure 3: Waveform — Holdout metrics per trial (Gini)

2.2.2 10-Fold Cross-Validation

Table 4: Waveform — 10-Fold CV Results (Gini)

Fold	**Accuracy**	**Precision**	**F-measure**
Fold 1	0.7440	0.7467	0.7438
Fold 2	0.7440	0.7444	0.7437
Fold 3	0.7680	0.7698	0.7682
Fold 4	0.7440	0.7448	0.7441
Fold 5	0.7340	0.7362	0.7344
Fold 6	0.7560	0.7555	0.7552
Fold 7	0.7720	0.7729	0.7713
Fold 8	0.7700	0.7704	0.7700
Fold 9	0.7180	0.7189	0.7179
Fold 10	0.7080	0.7101	0.7085
Average	**0.7458**	**0.7470**	**0.7457**

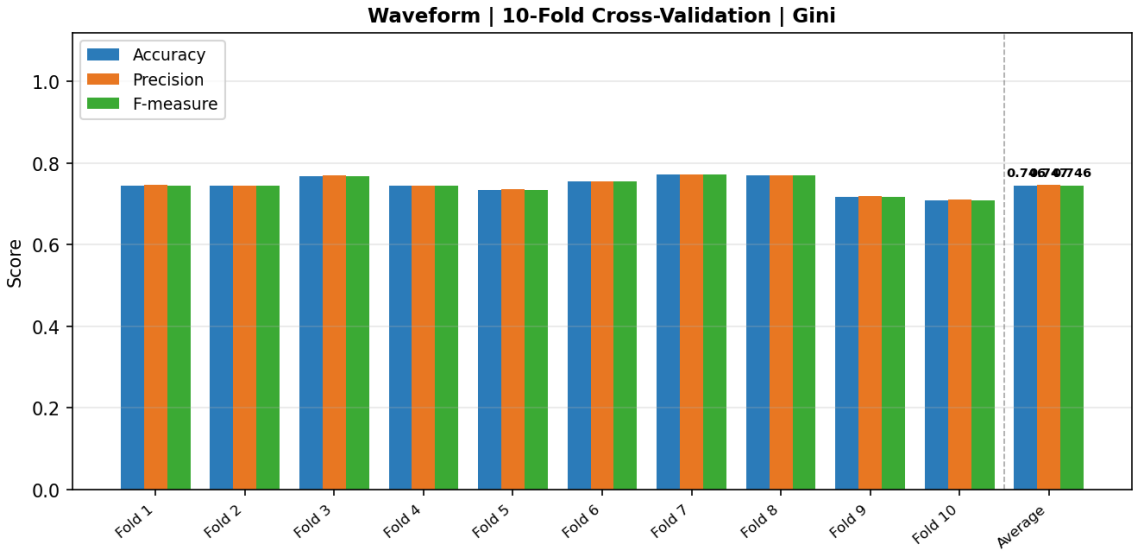


Figure 4: Waveform — 10-Fold CV metrics per fold (Gini)

3 Part II — Decision Tree with Entropy Impurity

3.1 German Credit Dataset

3.1.1 Holdout Method (90/10, 5 Trials)

Table 5: German Credit — Holdout Results (Entropy)

Trial	Accuracy	Precision	F-measure
Trial 1	0.7000	0.7329	0.7122
Trial 2	0.6400	0.6609	0.6483
Trial 3	0.6500	0.6716	0.6578
Trial 4	0.6800	0.6749	0.6772
Trial 5	0.6600	0.6600	0.6600
Average	**0.6660**	**0.6801**	**0.6711**

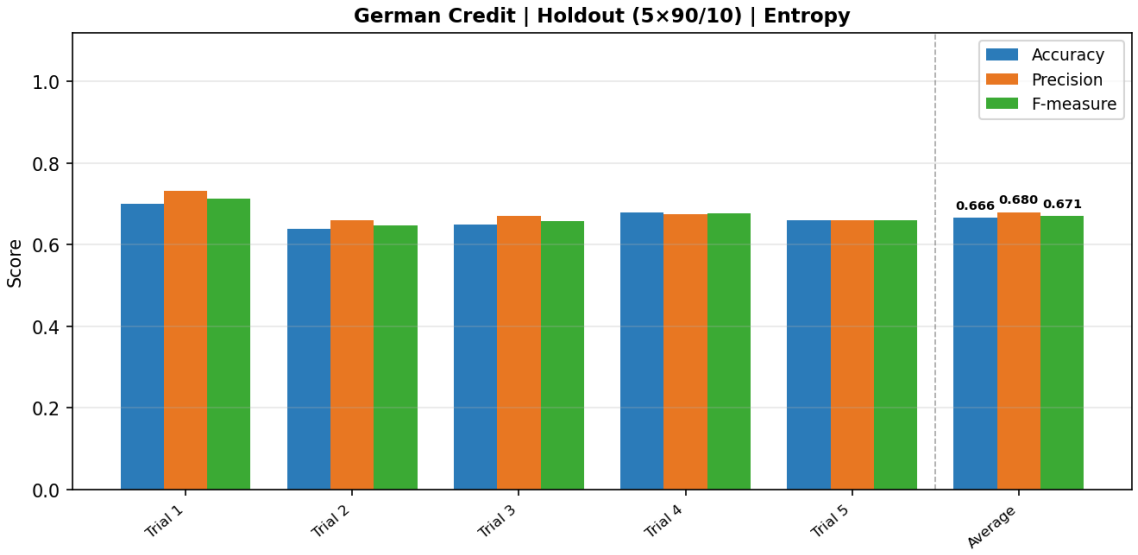


Figure 5: German Credit — Holdout metrics per trial (Entropy)

3.1.2 10-Fold Cross-Validation

Table 6: German Credit — 10-Fold CV Results (Entropy)

Fold	**Accuracy**	**Precision**	**F-measure**
Fold 1	0.7000	0.7329	0.7122
Fold 2	0.6400	0.6400	0.6400
Fold 3	0.7700	0.7632	0.7658
Fold 4	0.7500	0.7486	0.7492
Fold 5	0.7500	0.7342	0.7379
Fold 6	0.6200	0.6626	0.6357
Fold 7	0.6900	0.6877	0.6888
Fold 8	0.6200	0.6626	0.6357
Fold 9	0.6900	0.6869	0.6884
Fold 10	0.7000	0.6924	0.6950
Average	**0.6930**	**0.7011**	**0.6949**

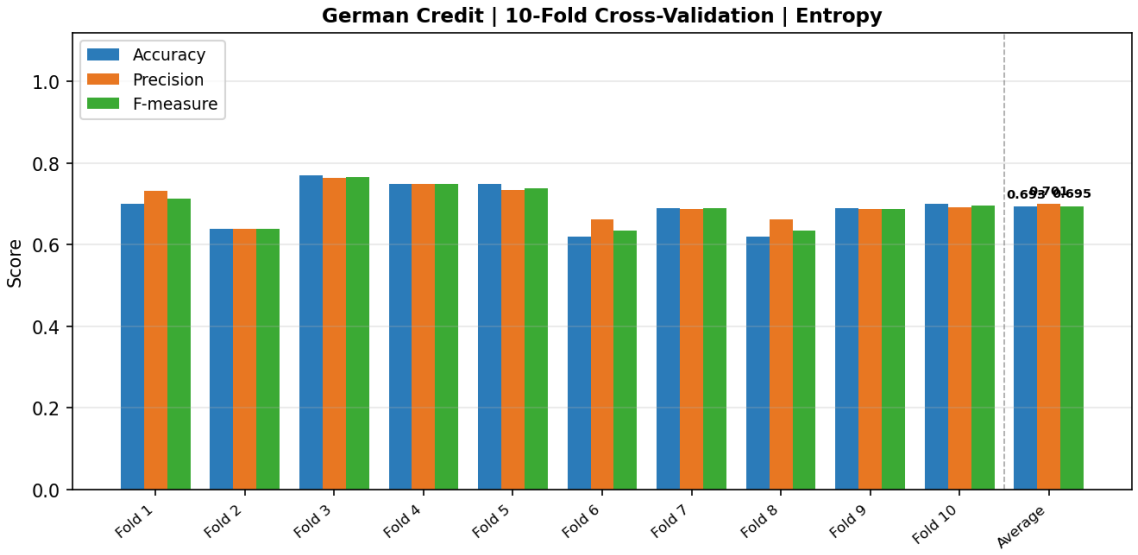


Figure 6: German Credit — 10-Fold CV metrics per fold (Entropy)

3.2 Waveform Dataset

3.2.1 Holdout Method (90/10, 5 Trials)

Table 7: Waveform — Holdout Results (Entropy)

Trial	Accuracy	Precision	F-measure
Trial 1	0.7360	0.7392	0.7359
Trial 2	0.7440	0.7468	0.7445
Trial 3	0.7320	0.7363	0.7330
Trial 4	0.7920	0.7919	0.7919
Trial 5	0.7400	0.7409	0.7401
Average	**0.7488**	**0.7510**	**0.7491**

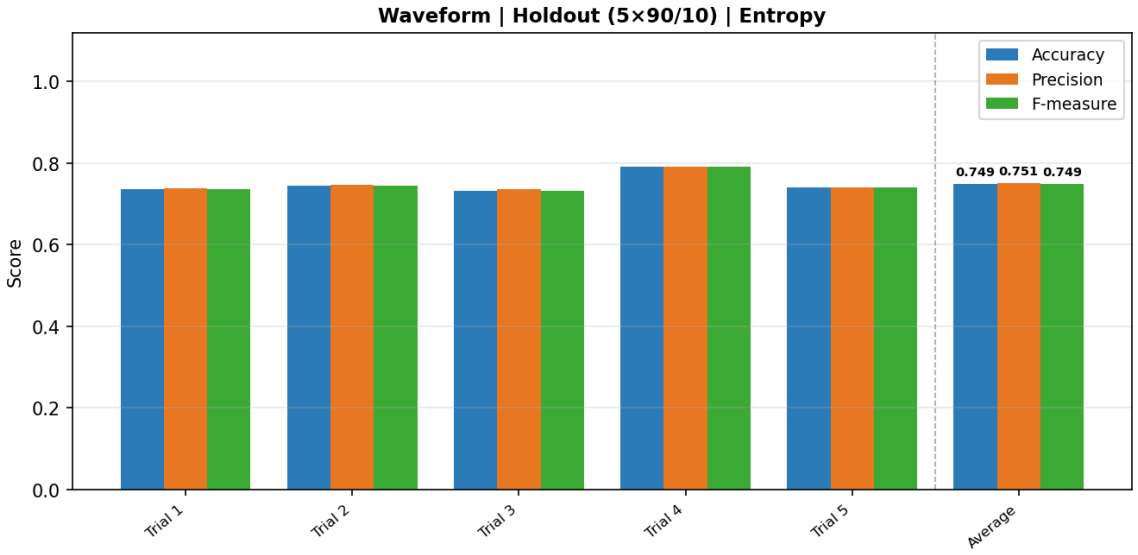


Figure 7: Waveform — Holdout metrics per trial (Entropy)

3.2.2 10-Fold Cross-Validation

Table 8: Waveform — 10-Fold CV Results (Entropy)

Fold	**Accuracy**	**Precision**	**F-measure**
Fold 1	0.7360	0.7392	0.7359
Fold 2	0.7780	0.7778	0.7775
Fold 3	0.7460	0.7488	0.7469
Fold 4	0.7520	0.7513	0.7514
Fold 5	0.7680	0.7681	0.7680
Fold 6	0.7640	0.7640	0.7637
Fold 7	0.7540	0.7537	0.7537
Fold 8	0.7520	0.7520	0.7520
Fold 9	0.7720	0.7717	0.7717
Fold 10	0.7580	0.7591	0.7578
Average	**0.7580**	**0.7586**	**0.7579**

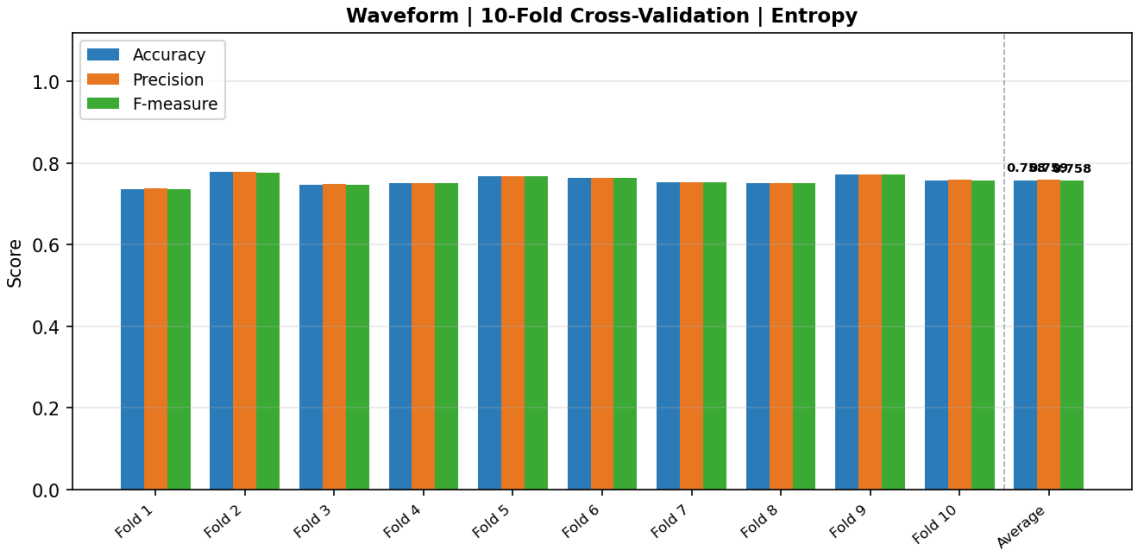


Figure 8: Waveform — 10-Fold CV metrics per fold (Entropy)

3.3 Comparison: Gini vs. Entropy (CV Accuracy)

Table 9 summarises the average 10-fold CV accuracy for both impurity measures across both datasets.

Table 9: Average 10-Fold CV Accuracy — Gini vs. Entropy

Dataset	Gini	Entropy
German Credit	0.6800	0.6930
Waveform	0.7458	0.7580

Figures 9 and 10 show the fold-by-fold accuracy and an aggregate bar comparison.

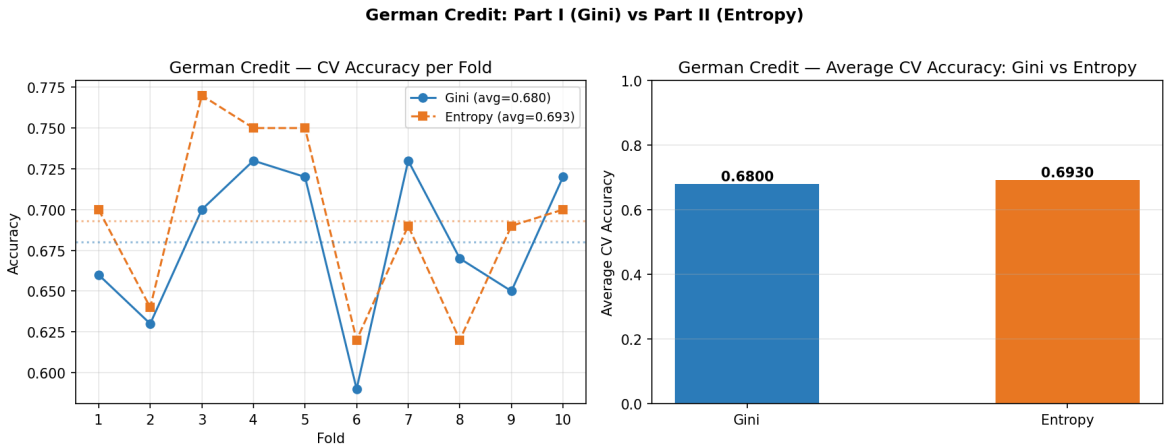


Figure 9: German Credit — CV accuracy per fold: Gini vs. Entropy (left) and average comparison bar chart (right).

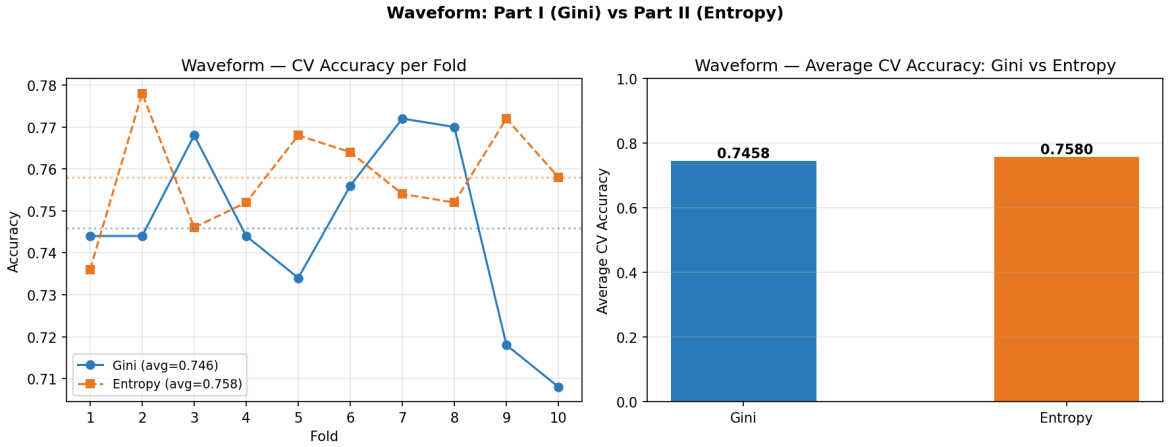


Figure 10: Waveform — CV accuracy per fold: Gini vs. Entropy (left) and average comparison bar chart (right).

3.3.1 Discussion

Entropy achieves marginally higher average CV accuracy on both datasets:

- **German Credit**: Gini = 0.680, Entropy = 0.693 (difference = +0.013).
- **Waveform**: Gini = 0.746, Entropy = 0.758 (difference = +0.012).

The differences are small, consistent with empirical literature showing that Gini and Information Gain (Entropy) select the same split in the vast majority of cases [1]. Entropy is slightly more expensive computationally (requires a logarithm) but the cost is negligible at these dataset sizes. The marginal improvement of Entropy may be attributed to its tendency to create more balanced splits, which can help on class-imbalanced tasks such as German Credit.

4 Part III — Effect of Tree Depth on Accuracy

4.1 Setup

Using a single 90/10 holdout split with `criterion='gini'`, the `max_depth` parameter is varied over {2, 5, 8, 11, 14, 17, 20, 23, 26, 29, 32, 35, 38, 41, 44, 47, 50}. For each depth, both training accuracy (on the 90 % partition) and test accuracy (on the 10 % partition) are recorded.

4.2 German Credit

Table 10: German Credit — Training & Test Accuracy vs. Max Depth (Gini)

Max Depth	Train Accuracy	Test Accuracy
2	0.7311	0.7300
5	0.8067	**0.7400**
8	0.9100	0.6900
11	0.9689	0.6700
14	0.9944	0.6300
17	1.0000	0.6600
20	1.0000	0.6600
23	1.0000	0.6600
26	1.0000	0.6600
29	1.0000	0.6600
32	1.0000	0.6600
35	1.0000	0.6600
38	1.0000	0.6600
41	1.0000	0.6600
44	1.0000	0.6600
47	1.0000	0.6600
50	1.0000	0.6600

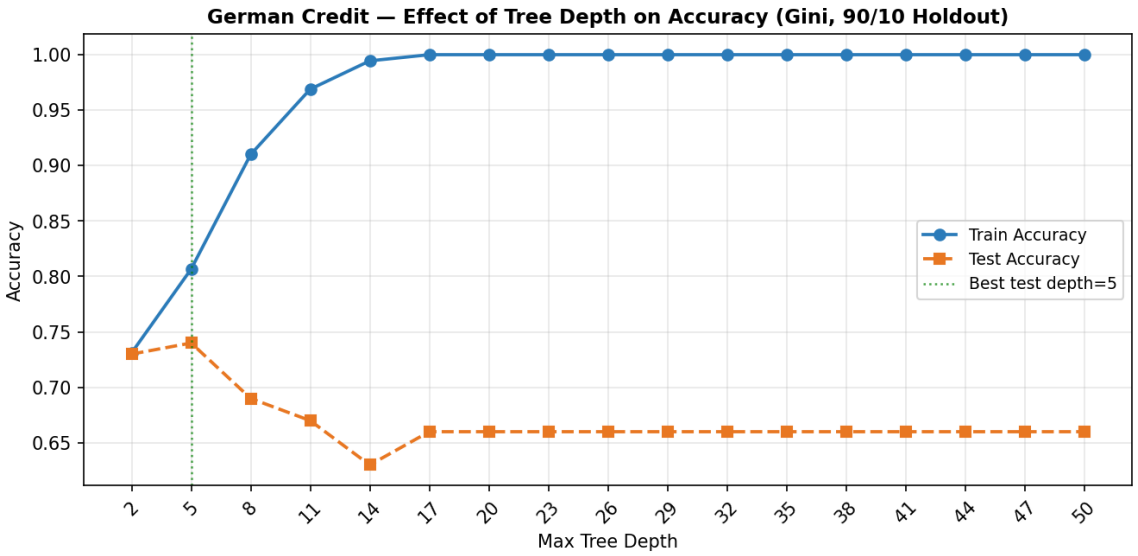


Figure 11: German Credit — Training vs. Test Accuracy as a function of Max Tree Depth (Gini, single 90/10 holdout).

4.3 Waveform

Table 11: Waveform — Training & Test Accuracy vs. Max Depth (Gini)

Max Depth	Train Accuracy	Test Accuracy
2	0.7031	0.7180
5	0.7996	0.7700
8	0.9084	**0.7940**
11	0.9678	0.7720
14	0.9918	0.7800
17	0.9993	0.7480
20	1.0000	0.7440
23	1.0000	0.7440
26	1.0000	0.7440
29	1.0000	0.7440
32	1.0000	0.7440
35	1.0000	0.7440
38	1.0000	0.7440
41	1.0000	0.7440
44	1.0000	0.7440
47	1.0000	0.7440
50	1.0000	0.7440

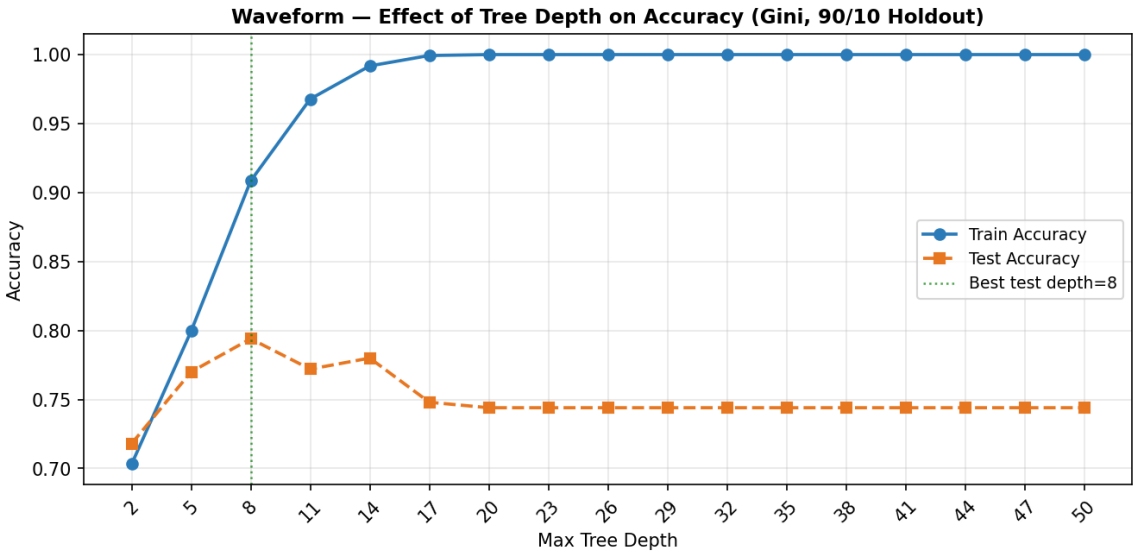


Figure 12: Waveform — Training vs. Test Accuracy as a function of Max Tree Depth (Gini, single 90/10 holdout).

4.4 Observations and Discussion

The depth plots reveal a textbook **bias–variance trade-off**:

Training accuracy increases monotonically with depth in both datasets, reaching 100 % once the tree is deep enough to perfectly memorise the training set. For German

Credit, saturation occurs around depth 17; for Waveform, around depth 20.

Test accuracy — German Credit. Test accuracy peaks early (**depth = 5**, test accuracy = 0.740) and then *decreases* as the tree grows deeper, stabilising at 0.660 for depths ≥ 17. This is a clear sign of overfitting: the increasingly complex tree fits training noise and fails to generalise. The optimal depth lies in the range [2, 5].

Test accuracy — Waveform. Test accuracy also peaks early (**depth = 8**, test accuracy = 0.794), then declines and stabilises at 0.744 for depths ≥ 20. The Waveform task has more inherent complexity (three classes, additive noise), so a slightly deeper tree is needed before the overfitting regime begins.

Key observation. There is a clear *gap* between training and test accuracy that widens as depth increases. Once the tree saturates to 100 % training accuracy, the test accuracy is already well below its maximum, demonstrating the danger of unconstrained tree growth.

Practical recommendation. These results strongly motivate *pre-pruning* (limiting `max_depth`) or *post-pruning* (cost-complexity pruning, `ccp_alpha` in scikit-learn). Ensemble methods such as Random Forests mitigate this variance issue by averaging many shallow trees.

Table 12: Summary: Best Test Accuracy and Corresponding Depth

Dataset	Best Depth	Best Test Acc.	Saturated Test Acc.
German Credit	5	0.740	0.660
Waveform	8	0.794	0.744

5 Summary and Conclusions

- **Holdout vs. Cross-Validation.** CV consistently yields higher and more stable accuracy estimates than a single holdout trial because all data points are used as test examples across folds. Variance between holdout trials (e.g., 0.630–0.700 for German Credit/Gini) illustrates the instability of a single 10 % test set at this sample size.
- **Gini vs. Entropy.** Entropy achieves marginally better accuracy (+1–1.5 percentage points) on both datasets. The computational overhead of Entropy is negligible at these scales, making it the preferred choice.
- **Tree Depth.** Full-depth trees overfit severely. Depth pruning at 5 (German Credit) and 8 (Waveform) recovers the best generalisation. Depths beyond 17–20 add no test accuracy benefit on either dataset.
- **Dataset Characteristics.** The Waveform dataset (5 000 samples, 3 balanced classes) supports higher accuracy ($\approx$ 75–80 %) compared with German Credit ($\approx$ 65–74 %), which has fewer samples, a harder decision boundary, and class imbalance (700:300).

References

[1] L. Breiman, J. Friedman, R. Olshen, and C. Stone, *Classification and Regression Trees*. Wadsworth, Belmont, CA, 1984.

[2] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[3] UCI Machine Learning Repository, *Statlog (German Credit Data)*, 1994.

[4] UCI Machine Learning Repository, *Waveform Database Generator (Version 1)*, 1988.